

Those questions highlighted in yellow are flagged out in the examiner's report as challenging.

- 1 The function f is defined by $f(z) = az^3 + bz^2 + cz + d$, where a, b, c and d are real numbers. Given that $2+i$ and -3 are roots of $f(z) = 0$, find b, c and d in terms of a . [4]

Qn	Solutions	Comments															
1	$f(2+i)=(2+i)^3 a+(2+i)^2 b+(2+i)c+d=0$ $(2+11i)a+(3+4i)b+(2+i)c+d=0$ $(2a+3b+2c+d)+(11a+4b+c)i=0$ Comparing real and imaginary parts, $2a+3b+2c+d=0$ and $11a+4b+c=0$. Also, $f(-3)=-27a+9b-3c+d=0$. $\text{Rewriting the three equations as } \begin{cases} 3b+2c+d+2a=0 \\ 4b+c+11a=0 \\ 9b-3c+d-27a=0 \end{cases},$ from the GC, $b=-a$, $c=-7a$ and $d=15a$. Alternative Since the coefficients a, b, c and d are real, $2+i$ is a root will imply that $2-i$ is also a root to $f(z)=0$. $\begin{aligned} az^3+bz^2+cz+d &= a[z-(2+i)][z-(2-i)](z-(-3)) \\ &= a[(z-2)+i][(z-2)-i](z+3) \\ &= a[(z-2)^2-i^2](z+3) \\ &= a(z^2-4z+5)(z+3) \\ &= a(z^3-z^2-7z+15) \end{aligned}$ Comparing coefficients, $b=-a$, $c=-7a$, and $d=15a$.	You can use the GC to evaluate e.g. $(2+i)^3$ Since we need b, c, and d in terms of a, better to let a be x_4 in the GC. SYSTEM MATRIX (3 × 5) <table><tr><td>3</td><td>2</td><td>1</td><td>2</td><td>0</td></tr><tr><td>4</td><td>1</td><td>0</td><td>11</td><td>0</td></tr><tr><td>9</td><td>-3</td><td>1</td><td>-27</td><td>0</td></tr></table> [SYSM](3,5)=0 SOLUTION SET $\begin{aligned} x_1 &= 0-1x_4 \\ x_2 &= 0-7x_4 \\ x_3 &= 0+15x_4 \\ x_4 &= x_4 \end{aligned}$ Alternative Note the condition: a, b, c and d are real numbers. Hence we can apply The Fundamental Theorem of Algebra $z=2+i$ is a root to the equation, $(z-(2+i))$ is a factor to the expression. Learn how to expand $[z-(2+i)][z-(2-i)]$ effectively.	3	2	1	2	0	4	1	0	11	0	9	-3	1	-27	0
3	2	1	2	0													
4	1	0	11	0													
9	-3	1	-27	0													

2 The curve C has equation $y = x^3 + x - 1$.

(i) C crosses the x -axis at the point with coordinates $(a, 0)$. Find the value of a correct to 3 decimal places. [1]

(ii) You are given that $b > a$.

The region P is bounded by C , the x -axis and the lines $x = -1$ and $x = 0$. The region Q is bounded by C , the line $x = b$ and the part of the x -axis between $x = a$ and $x = b$. Given that the area of Q is 2 times the area of P , find the value of b correct to 3 decimal places. [4]

Qn	Solutions	Comments
2i	When $x = a$, $y = 0$ $a^3 + a - 1 = 0$ By GC, $a \approx 0.68233$ $a \approx 0.682$ (to 3 d.p.)	Read the question, no need to solve for a algebraically.
2ii	$\int_a^b x^3 + x - 1 \, dx = 2 \int_{-1}^0 -(x^3 + x - 1) \, dx$ $\left[\frac{1}{4}x^4 + \frac{1}{2}x^2 - x \right]_{0.68233}^b = 2(1.75)$ $\frac{1}{4}b^4 + \frac{1}{2}b^2 - b - (-0.39535) = 3.5$ $\frac{1}{4}b^4 + \frac{1}{2}b^2 - b - 3.1046 = 0$ From the GC, $b \approx 1.892$ (to 3 d.p.) since $b > a > 0$.	Note that region P falls below the x axis, hence area of $P = \int_{-1}^0 -y \, dx$ Since non-exact answer is required, try to use the GC wherever you can. Note that to avoid rounding off error, you should use $a = 0.68233$ (i.e. at least 5 dp) Read question carefully, it is mentioned that $b > a$ and $a > 0$.

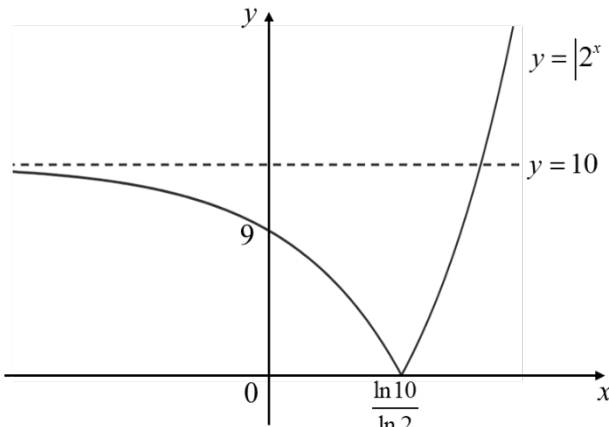
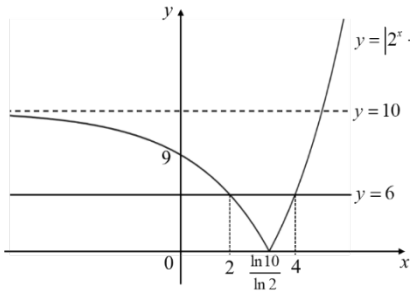
3 A function is defined as $f(x) = 2x^3 - 6x^2 + 6x - 12$.

(i) Show that $f(x)$ can be written in the form $p\{(x+q)^3 + r\}$, where p , q and r are constants to be found. [2]

(ii) Hence, or otherwise, describe a sequence of transformations that transform the graph of $y = x^3$ onto the graph of $y = f(x)$. [3]

Qn	Solutions	Comments
3i	$f(x) = 2x^3 - 6x^2 + 6x - 12 = 2(x^3 - 3x^2 + 3x - 6)$ $= 2(x^3 - 3x^2 + 3x - 1 - 5)$ $= 2\{(x-1)^3 - 5\}$ $p = 2, q = -1, r = -5$ Alternative $f(x) = p\{(x+q)^3 + r\} = p(x^3 + 3qx^2 + 3q^2x + q^3 + r)$ $2x^3 - 6x^2 + 6x - 12 = p(x^3 + 3qx^2 + 3q^2x + q^3 + r)$ Comparing coefficients, $p = 2$, $3pq = -6$, $p(q^3 + r) = -12$. Solving the equations in order gives $p = 2$, $q = -1$, $r = -5$. $\therefore f(x) = 2\{(x-1)^3 - 5\}$	Recall binomial expansion $(x+a)^3$ $= x^3 + 3ax^2 + 3a^2x + a^3$ By observation, $a = -1$
3ii	$y = x^3 \rightarrow y = (x-1)^3 \rightarrow y = (x-1)^3 - 5 \rightarrow y = 2\{(x-1)^3 - 5\}$ 1) Translation of 1 unit in the positive x-direction, followed by 2) Translation of 5 units in the negative y-direction, followed by 3) Stretch parallel to the y-axis by factor 2, with x-axis invariant.	Alternative (2) $\rightarrow$ (3) $\rightarrow$ (1)

- 4 (i) Sketch the graph of $y = |2^x - 10|$, giving the exact values of any points where the curve meets the axes. [3]
- (ii) Without using a calculator, and showing all your working, find the exact interval, or intervals, for which $|2^x - 10| \leq 6$. Give your answer in its simplest form. [3]

Qn	Solutions	Comments
4i	 The graph shows the function $y = 2^x - 10$ on a Cartesian coordinate system. The curve has a minimum point on the x-axis at $x = \frac{\ln 10}{\ln 2}$. A horizontal dashed line at $y = 10$ represents the asymptote as $x \rightarrow -\infty$. The y-intercept is at $(0, 9)$.	Read the question, need exact values for axial intercepts Note that as $x \rightarrow -\infty$, $2^x \rightarrow 0$. Hence $y = 2^x - 10 \rightarrow 10$ So $y = 10$ is a horizontal asymptote.
4ii	When $2^x - 10 = 6$, $2^x = 16$ $x = 4$. When $-(2^x - 10) = 6$, $2^x = 4$ $x = 2$. Interval for which $2^x - 10 \leq 6$ is $[2, 4]$.  The second graph shows the same function $y = 2^x - 10$ but with a horizontal line at $y = 6$ drawn. The region where the curve is below this line is shaded, corresponding to the interval $x \in [2, 4]$. The x-axis is marked with 0, 2, $\frac{\ln 10}{\ln 2}$, and 4.	We need to find the x-coordinates of the points of intersections algebraically. Thus need to drop the modulus We want the curve ($y = 2^x - 10$) to be below the straight line ($y = 6$). You can leave your answer as $2 \leq x \leq 4$

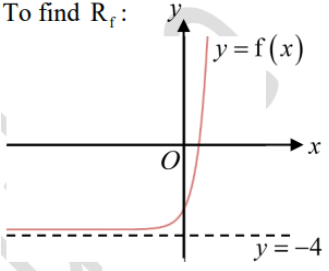
5 The functions f and g are defined by

$$f(x) = e^{2x} - 4, \quad x \in \mathbb{R},$$

$$g(x) = x + 2, \quad x \in \mathbb{R}.$$

(i) Find $f^{-1}(x)$ and state its domain. [3]

(ii) Find the exact solution of $fg(x) = 5$, giving your answer in its simplest form. [3]

Qn	Solutions	Comments
5i	Let $y = f(x)$. $y = e^{2x} - 4$ $e^{2x} = y + 4$ $x = \frac{1}{2} \ln(y + 4)$ $\therefore f^{-1}(x) = \frac{1}{2} \ln(x + 4), \quad x > -4; \quad D_{f^{-1}} = R_f = (-4, \infty)$	To find R_f: 
5ii	$fg(x) = 5$ $f^{-1}fg(x) = f^{-1}(5)$ $g(x) = \frac{1}{2} \ln(5 + 4)$ $x + 2 = \ln 3$ $x = -2 + \ln 3$ Alternative $fg(x) = 5$ $f(x + 2) = 5$ $e^{2(x+2)} - 4 = 5$ $e^{2(x+2)} = 9$ $2(x + 2) = \ln 9$ $x = -2 + \ln 3$	Recall that $a = f(b)$ $\Leftrightarrow f^{-1}(a) = f^{-1}f(b) = b$ Since $f^{-1}f(x) = x$

6 (i) By writing $\frac{1}{4r^2-1}$ in partial fractions, find an expression for $\sum_{r=1}^n \frac{1}{4r^2-1}$. [4]

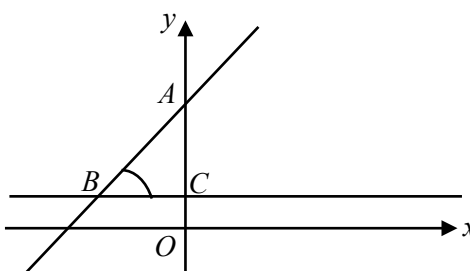
(ii) Hence find the exact value of $\sum_{r=1}^{\infty} \frac{1}{4r^2-1}$. [2]

Qn	Solutions	Comments
6i	$\frac{1}{4r^2-1} = \frac{A}{2r-1} + \frac{B}{2r+1} = \frac{1}{2(2r-1)} - \frac{1}{2(2r+1)}$ $\sum_{r=1}^n \frac{1}{4r^2-1} = \sum_{r=1}^n \left(\frac{1}{2(2r-1)} - \frac{1}{2(2r+1)} \right)$ $= \frac{1}{2} \left[\frac{1}{1} - \frac{1}{3} + \frac{1}{3} - \frac{1}{5} + \dots + \frac{1}{2n-3} - \frac{1}{2n-1} + \frac{1}{2n-1} - \frac{1}{2n+1} \right]$ $= \frac{1}{2} \left(1 - \frac{1}{2n+1} \right)$	Revise how to do partial fractions: $\frac{1}{4r^2-1} = \frac{1}{(2r-1)(2r+1)}$ $= \frac{A}{2r-1} + \frac{B}{2r+1}$ $1 = A(2r+1) + B(2r-1)$ When $r = -\frac{1}{2}$, $B = -\frac{1}{2}$ When $r = \frac{1}{2}$, $A = \frac{1}{2}$ $\sum_{r=1}^n \frac{1}{4r^2-1}$ is actually in terms of n, not r.
6ii	$\sum_{r=1}^{\infty} \frac{1}{4r^2-1} = \sum_{r=1}^{\infty} \frac{1}{4r^2-1} - \sum_{r=1}^{10} \frac{1}{4r^2-1}$ $= \lim_{n \rightarrow \infty} \left[\frac{1}{2} \left(1 - \frac{1}{2n+1} \right) \right] - \frac{1}{2} \left[1 - \frac{1}{2(10)+1} \right]$ $= \frac{1}{2} - \frac{10}{21}$ $= \frac{1}{42}$	Note: - you can use (i) <u>only when the index starts from $r=1$</u>. - As $n \rightarrow \infty$, $\frac{1}{2n+1} \rightarrow 0$ Pay attention to the presentation. The following are correct ✓ $\sum_{r=1}^{\infty} u_r = \lim_{n \rightarrow \infty} \sum_{r=1}^n u_r$ ✓ $\sum_{r=1}^{\infty} u_r = \frac{1}{2}$ ✓ $\sum_{r=1}^n u_r \rightarrow \frac{1}{2}$ as $n \rightarrow \infty$

7 A curve C has equation $y = xe^{-x}$.

(i) Find the equations of the tangents to C at the points where $x = 1$ and $x = -1$. [6]

(ii) Find the acute angle between these tangents. [2]

Qn	Solutions	Comments
7i	$y = xe^{-x}$ $\frac{dy}{dx} = e^{-x} + x(-e^{-x})$ $= e^{-x}(1 - x)$ At $x = 1$, $\frac{dy}{dx} = 0$ and $y = \frac{1}{e}$. Equation of tangent: $y = \frac{1}{e}$ At $x = -1$, $\frac{dy}{dx} = 2e$ and $y = -e$. Equation of tangent: $y + e = 2e(x + 1)$ $y = 2ex + e$	When $x = 1$, $\frac{dy}{dx} = 0 \Rightarrow$ tangent is parallel to x-axis i.e. tangent is horizontal line with equation $y = \frac{1}{e}$
7ii	Acute angle between tangents $= \tan^{-1}(2e)$ $= 79.6^\circ \text{ (to 1 d.p.)}$	 Angle between tangents = $\angle ABC$ $\angle ABC = \tan^{-1}\left(\frac{AC}{BC}\right)$ $= \tan^{-1}\left(\frac{\text{rise}}{\text{run}}\right)$ $= \tan^{-1}(\text{gradient of st. line})$

- 8** (a) An arithmetic series has first term a and common difference $2a$, where $a \neq 0$. A geometric series has first term a and common ratio 2. The k th term of the geometric series is equal to the sum of the first 64 terms of the arithmetic series. Find the value of k . [3]
- (b) A geometric series has first term f and common ratio r , where $f, r \in \mathbb{R}$ and $f \neq 0$. The sum of the first four terms of the series is 0. Find the possible values of f and r . Find also, in terms of f , the possible values of the sum of the first n terms of the series. [4]
- (c) The first term of an arithmetic series is negative. The sum of the first four terms of the series is 14 and the product of the first four terms of the series is 0. Find the 11th term of the series. [4]

Qn	Solutions	Comments
8a	$a(2)^{k-1} = \frac{64}{2} [2a + (64-1)(2a)]$ $a2^{k-1} = 32(128a)$ $2^{k-1} = 2^5 (2^7) \quad (\text{Since } a \neq 0)$ $k-1 = 12$ $k = 13$	
8b	Given: $S_4 = 0$ $f + fr + fr^2 + fr^3 = 0$ $1 + r + r^2 + r^3 = 0 \quad (\because f \neq 0)$ $(r+1)(r^2+1) = 0$ $r = -1 \text{ or } r^2 = -1 \quad (\text{reject})$ <u>Alternative</u> Since $S_4 = 0$, hence $r < 0$ $\frac{f(1-r^4)}{1-r} = 0$ $1-r^4 = 0 \quad (\because f \neq 0)$ $r = -1 \quad (\because r < 0)$ Hence, $f \in \mathbb{R} \setminus \{0\}$, $r = -1$. $\text{Sum of first } n \text{ terms} = \frac{f(1-(-1)^n)}{1-(-1)} = \frac{f}{2} [1-(-1)^n]$ $= \begin{cases} f, & \text{if } n \text{ is odd} \\ 0, & \text{if } n \text{ is even} \end{cases}$	If $r = 1$, then $S_4 = f + f + f + f = 4f \neq 0$ After solving for $r = -1$, note that the first four terms are $f, -f, f, -f$ respectively, and their sum is always 0. Hence, there are no other restrictions on f, other than it not being 0. So $f \in \mathbb{R} \setminus \{0\}$. Observe that when n is odd, $(-1)^n = -1$; n is even, $(-1)^n = 1$.
8c	Let the arithmetic series have first term u and common difference d. $\frac{4}{2}(2u+3d) = 14 \Rightarrow 2u+3d = 7 \quad \text{----- (1)}$ and $u(u+d)(u+2d)(u+3d) = 0$	Read the question carefully, we are told that $u < 0$.

$u = 0$ or $u = -d$ or $u = -2d$ or $u = -3d$ ----- (2) [If $u = -2d$, eqn (1) gives $u = 14 > 0$.] [If $u = -3d$, eqn (1) gives $u = 7 > 0$.] Since $u < 0$, we must have $u = -d$. Substituting into (1): $d = 7$ Hence, $u = -7$. 11^{th} term $= -7 + 10(7) = 63$	We should not expand and obtain a quartic equation in terms of u .
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9 (i) The complex number w can be expressed as $\cos \theta + i \sin \theta$.

(a) Show that $w + \frac{1}{w}$ is a real number. [2]

(b) Show that $\frac{w-1}{w+1}$ can be expressed as $k \tan \frac{1}{2} \theta$, where k is a complex number to be found. [4]

(ii) The complex number z has modulus 1. Find the modulus of the complex number $\frac{z-3i}{1+3iz}$. [5]

Qn	Solutions	Comments
9ia	$w + \frac{1}{w} = \cos \theta + i \sin \theta + \frac{1}{\cos \theta + i \sin \theta}$ $= \cos \theta + i \sin \theta + \frac{\cos \theta - i \sin \theta}{(\cos \theta + i \sin \theta)(\cos \theta - i \sin \theta)}$ $= \cos \theta + i \sin \theta + \frac{\cos \theta - i \sin \theta}{\cos^2 \theta + \sin^2 \theta}$ $= \cos \theta + i \sin \theta + \frac{\cos \theta - i \sin \theta}{1}$ $= 2 \cos \theta$ Hence $w + \frac{1}{w}$ is a real number. Alternative $w = \cos \theta + i \sin \theta = e^{i\theta}$ $w + \frac{1}{w} = e^{i\theta} + \frac{1}{e^{i\theta}}$ $= e^{i\theta} + e^{-i\theta} \quad \text{---} (*)$ $= \cos \theta + i \sin \theta + \cos(-\theta) + i \sin(-\theta)$ $= \cos \theta + i \sin \theta + \cos \theta - i \sin \theta$ $= 2 \cos \theta$ Hence $w + \frac{1}{w}$ is a real number.	To show $w + \frac{1}{w}$ is a real number, we need to get $w + \frac{1}{w} = a + (0)i$. In other words, find out what $w + \frac{1}{w}$ is, in polar trigo form and see that the imaginary part is 0. The following are useful identities to note * $e^{i\theta} + e^{-i\theta} = 2 \cos \theta$ * $e^{i\theta} - e^{-i\theta} = 2i \sin \theta$
9ib	$\frac{w-1}{w+1} = \frac{\cos \theta - 1 + i \sin \theta}{\cos \theta + 1 + i \sin \theta}$ $= \frac{(\cos \theta - 1 + i \sin \theta)(\cos \theta + 1 - i \sin \theta)}{(\cos \theta + 1 + i \sin \theta)(\cos \theta + 1 - i \sin \theta)}$ $= \frac{\cos^2 \theta - 1 + \sin^2 \theta + 2i \sin \theta}{\cos^2 \theta + 2 \cos \theta + 1 + \sin^2 \theta}$ $= \frac{2i \sin \theta}{2 + 2 \cos \theta} = \frac{i \sin \theta}{1 + \cos \theta} = \frac{i \left(2 \sin \frac{\theta}{2} \cos \frac{\theta}{2} \right)}{2 \cos^2 \frac{\theta}{2}} = i \tan \frac{\theta}{2}$ $k = i$	Note $z^* = \operatorname{Re}(z) - i \operatorname{Im}(z)$ Thus $(\cos \theta + 1 + i \sin \theta)^* = (\cos \theta + 1) - i \sin \theta$ Since we need to get $k \tan \frac{\theta}{2}$ and we are given $w = \cos \theta + i \sin \theta$, this suggest that we have to

	Step 1: evaluate $\frac{w-1}{w+1}$ in trigo form Step 2: apply double angle formula Alternative $\frac{w-1}{w+1} = \frac{e^{i\theta} - 1}{e^{i\theta} + 1} = \frac{e^{\frac{i\theta}{2}} \left(e^{\frac{i\theta}{2}} - e^{-\frac{i\theta}{2}} \right)}{e^{\frac{i\theta}{2}} \left(e^{\frac{i\theta}{2}} + e^{-\frac{i\theta}{2}} \right)} = \frac{2i \sin \frac{\theta}{2}}{2 \cos \frac{\theta}{2}} = i \tan \frac{\theta}{2}$ $k = i$	This alternative method is a strategy that easily helps us obtain $\frac{\theta}{2}$ by using exponential form. We also applied the following results 1. $e^{i\theta} + e^{-i\theta} = 2 \cos \theta$ 2. $e^{i\theta} - e^{-i\theta} = 2i \sin \theta$ (1) is proven in (ia). You can prove (2) using the same method as in (ia).
9ii	Let $z = x + iy$, $x, y \in \mathbb{R}$. Given $z = 1$, $\sqrt{x^2 + y^2} = 1 \Rightarrow x^2 + y^2 = 1$. $\begin{aligned} \left \frac{z-3i}{1+3iz} \right &= \frac{ x + (y-3)i }{ 1 + 3i(x+iy) } \\ &= \frac{ x + (y-3)i }{ 1 - 3y + 3xi } \\ &= \frac{\sqrt{x^2 + (y-3)^2}}{\sqrt{(1-3y)^2 + (3x)^2}} \\ &= \frac{\sqrt{x^2 + y^2 - 6y + 9}}{\sqrt{1 - 6y + 9y^2 + 9x^2}} \\ &= \frac{\sqrt{10 - 6y}}{\sqrt{10 - 6y}} \quad (\because x^2 + y^2 = 1) \\ &= 1 \end{aligned}$ For tutors: $\begin{aligned} \left \frac{z-3i}{1+3iz} \right &= \left \frac{z-3i}{zz^* + 3iz} \right \quad (\text{since } z = 1) \\ &= \frac{1}{ z } \left \frac{z-3i}{z^* + 3i} \right \\ &= \left \frac{z-3i}{z^* + 3i} \right \quad (\text{since } z = 1) \\ &= \left \frac{w}{w^*} \right \quad (\text{where } w = z - 3i) \\ &= 1 \end{aligned}$	For tutors: Check students workings! Students have limited understanding of modulus, a very common mistake is $z - 3i = z - 3i$ Note that correct answers were obtained from wrong work, as can be seen in the example shown below $\begin{aligned} \left \frac{z-3i}{1+3iz} \right &= \frac{ z-3i }{ 1+3iz } \\ &= \frac{ z + -3i }{ 1 + 3i z } \\ &= \frac{4}{1+3} \\ &= 1 \end{aligned}$

10 A curve C has parametric equations

$$\begin{aligned}x &= a(2\cos\theta - \cos 2\theta), \\y &= a(2\sin\theta - \sin 2\theta),\end{aligned}$$

for $0 \leq \theta \leq 2\pi$.

(i) Sketch C and state the Cartesian equation of its line of symmetry. [2]

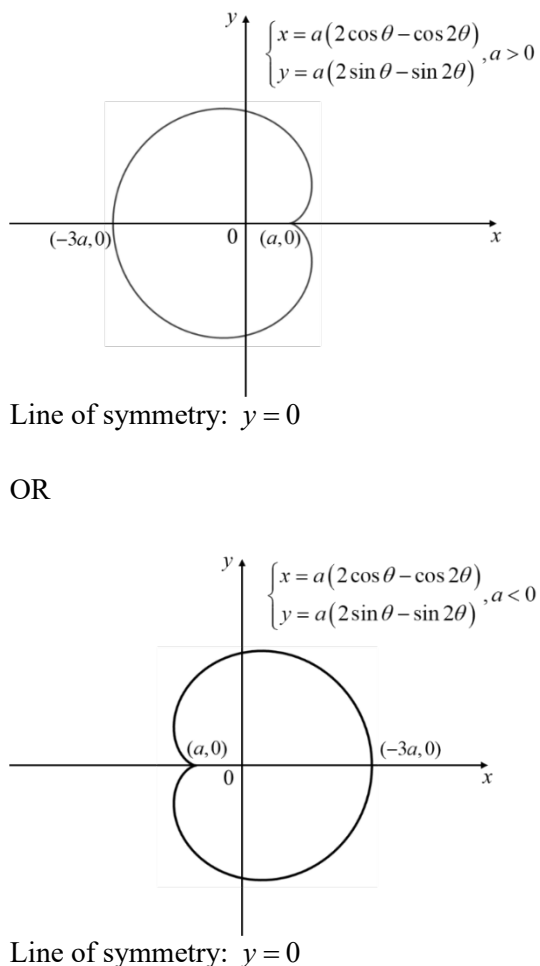
(ii) Find the values of θ at the points where C meets the x -axis. [2]

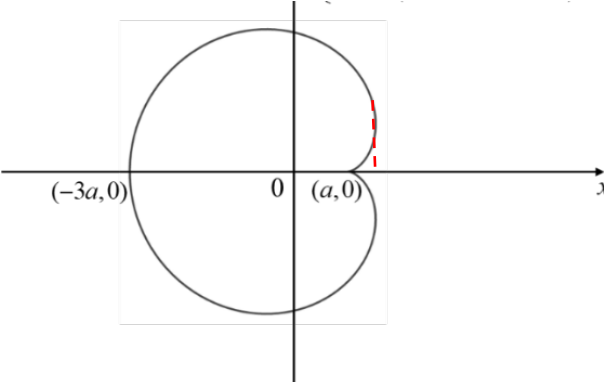
(iii) Show that the area enclosed by the x -axis and the part of C above the x -axis, is given by

$$\int_{\theta_1}^{\theta_2} a^2 (4\sin^2\theta - 6\sin\theta\sin 2\theta + 2\sin^2 2\theta) d\theta,$$

where θ_1 and θ_2 should be stated. [3]

(iv) Hence find, in terms of a , the exact total area enclosed by C . [5]

Qn	Solutions	Comments
10i	 Line of symmetry: $y = 0$ OR Line of symmetry: $y = 0$	Here, we shall assume $a \neq 0$, i.e. either $a > 0$ or $a < 0$ and sketch one of the graphs.
10ii	When $y = 0$,	This is secondary school work. Note $0 \leq \theta \leq 2\pi$.

	$2 \sin \theta - \sin 2\theta = 0$ $2 \sin \theta - 2 \sin \theta \cos \theta = 0$ $\sin \theta (1 - \cos \theta) = 0$ $\sin \theta = 0 \quad \text{or} \quad \cos \theta = 1$ $\theta = 0 \text{ or } \pi \text{ or } 2\pi \text{ or } \theta = 0 \text{ or } 2\pi$ Values of θ are 0, π, 2π.	
10iii	$\frac{dx}{d\theta} = a(-2 \sin \theta + 2 \sin 2\theta)$ When $x = -3a$, $\theta = \pi$. When $x = a$, $\theta = 0$.  Required area $= \int_{-3a}^{x_1} y \, dx - \int_a^{x_1} y \, dx$ $= \int_{-3a}^{x_1} y \, dx + \int_{x_1}^a y \, dx$ $= \int_{-3a}^a y \, dx$ $= \int_{\pi}^0 a(2 \sin \theta - \sin 2\theta) \cdot a(-2 \sin \theta + 2 \sin 2\theta) \, d\theta$ $= \int_{\pi}^0 -a^2 (2 \sin \theta - \sin 2\theta)(2 \sin \theta - 2 \sin 2\theta) \, d\theta$ $= \int_0^{\pi} a^2 (4 \sin^2 \theta - 6 \sin \theta \sin 2\theta + 2 \sin^2 2\theta) \, d\theta$ $\theta_1 = 0$ and $\theta_2 = \pi$	In this working, we assume that $a > 0$. If $a < 0$, you should obtain $\theta_1 = \pi$ and $\theta_2 = 2\pi$.
10iv	Total area enclosed by C $= 2 \int_0^{\pi} a^2 (4 \sin^2 \theta - 6 \sin \theta \sin 2\theta + 2 \sin^2 2\theta) \, d\theta$ $= 2a^2 \int_0^{\pi} 2(1 - \cos 2\theta) - 12 \cos \theta \sin^2 \theta + (1 - \cos 4\theta) \, d\theta$ $= 2a^2 \left[2\theta - \sin 2\theta - 4 \sin^3 \theta + \theta - \frac{1}{4} \sin 4\theta \right]_0^{\pi}$ $= 6a^2 \pi$	Remember to double the area in (iii) since the region enclosed by C is symmetrical about the x-axis

11 Scientists are investigating how the temperature of water changes in various environments.

(i) The scientists begin by investigating how hot water cools.

The water is heated in a container and then placed in a room which is kept at a constant temperature of 16°C . The temperature of the water t minutes after it is placed in the room is $\theta^{\circ}\text{C}$. The temperature decreases at a rate proportional to the difference between the temperature of the water and the temperature of the room. The temperature of the water falls from a value of 80°C to 32°C in the first 30 minutes.

(a) Write down a differential equation for this situation. Solve this differential equation to get θ as an exact function of t . [6]

(b) Find the temperature of the water 45 minutes after it is placed in the room. [1]

(ii) The scientists then model the thickness of ice on a pond.

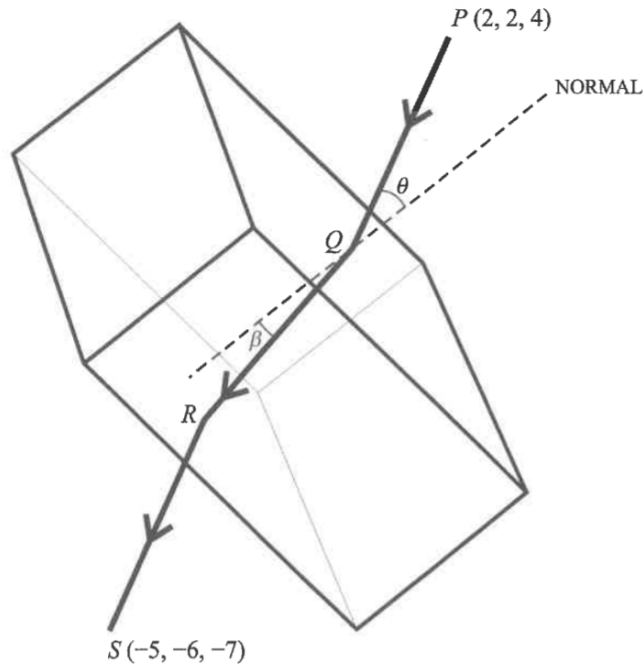
In winter the surface of the water in the pond freezes. Once the thickness of the ice reaches 3 cm, it is safe to skate on the ice. The thickness of the ice is T cm, t minutes after the water starts to freeze. The freezing of the water is modelled by a differential equation in which the rate of change of the thickness of the ice is inversely proportional to its thickness. It is given that $T = 0$ when $t = 0$. After 60 minutes the ice is 1 cm thick.

Find the time from when freezing commences until the ice is first safe to skate on. [6]

Qn	Solutions	Comments
11ia	$\frac{d\theta}{dt} = -k(\theta - 16), \quad k > 0$ $\int \frac{1}{\theta - 16} d\theta = \int -k dt$ $\ln \theta - 16 = -kt + C$ $ \theta - 16 = e^{-kt+C} = Be^{-kt}$ $\theta - 16 = Ae^{-kt}$ $\theta = 16 + Ae^{-kt}$ When $t = 0$, $\theta = 80$. $80 = 16 + A \Rightarrow A = 64$ $\theta = 16 + 64e^{-kt}$ When $t = 30$, $\theta = 32$. $32 = 16 + 64e^{-30k}$ $\Rightarrow e^{-30k} = \frac{1}{4}$ $\Rightarrow -30k = -\ln 4$ $\Rightarrow k = \frac{\ln 4}{30} = \frac{\ln 2}{15}$ $\therefore \theta = 16 + 64e^{-\frac{\ln 2}{15}t}$	Need to find θ in terms of t
11ib	At $t = 45$,	

	$\theta = 16 + 64e^{-\frac{45 \ln 2}{15}}$ $= 16 + 64e^{-3 \ln 2}$ $= 16 + 64e^{\ln \frac{1}{8}}$ $= 16 + 64\left(\frac{1}{8}\right)$ $= 24$ Temperature is 24°C.	
11ii	$\frac{dT}{dt} = \lambda \left(\frac{1}{T} \right), \lambda > 0$ $\int T \, dT = \int \lambda \, dt$ $\frac{1}{2} T^2 = \lambda t + D$ When $t = 0$, $T = 0$. So $D = 0$ and $\frac{1}{2} T^2 = \lambda t$. When $t = 60$, $T = 1$. $\frac{1}{2} = \lambda(60) \Rightarrow \lambda = \frac{1}{120}$ $\therefore T^2 = \frac{t}{60}$ When $T = 3$, $t = 3^2(60) = 540$. Ice is first safe to skate on 540 minutes after freezing commences.	Concept of inverse proportion is taught in secondary school Remember to give the units in your final answer

12



A ray of light passes from air into a material made into a rectangular prism. The ray of light is sent in direction $\begin{pmatrix} -2 \\ -3 \\ -6 \end{pmatrix}$ from a light source at the point P with coordinates $(2, 2, 4)$. The prism is placed so that the ray of light passes through the prism, entering at the point Q and emerging at the point R and is picked up by a sensor at point S with coordinates $(-5, -6, -7)$. The acute angle between PQ and the normal to the top of the prism at Q is θ and the acute angle between QR and the same normal is β (see diagram).

It is given that the top of the prism is a part of the plane $x + y + z = 1$, and the base of the prism is a part of the plane $x + y + z = -9$. It is also given that the ray of light along PQ is parallel to the ray of light along RS so that P, Q, R and S lie in the same plane.

- (i) Find the exact coordinates of Q and R . [5]
 - (ii) Find the values of $\cos \theta$ and $\cos \beta$. [3]
 - (iii) Find the thickness of the prism measured in the direction of the normal at Q . [3]
- Snell's law states that $\sin \theta = k \sin \beta$, where k is a constant called the refractive index.
- (iv) Find k for the material of this prism. [1]
 - (v) What can be said about the value of k for a material for which $\beta > \theta$? [1]

Qn	Solutions	Comments
12i	$l_{PQ} : \mathbf{r} = \begin{pmatrix} 2 \\ 2 \\ 4 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ -3 \\ -6 \end{pmatrix}, \lambda \in \mathbb{R} \text{ and } \pi_{top} : \mathbf{r} \cdot \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = 1$ To find Q, $\left[\begin{pmatrix} 2 \\ 2 \\ 4 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ -3 \\ -6 \end{pmatrix} \right] \cdot \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = 1$ $(2+2+4) + \lambda(-2-3-6) = 1 \Rightarrow \lambda = \frac{7}{11}$ $\therefore \mathbf{r} = \begin{pmatrix} 2 \\ 2 \\ 4 \end{pmatrix} + \frac{7}{11} \begin{pmatrix} -2 \\ -3 \\ -6 \end{pmatrix} = \frac{1}{11} \begin{pmatrix} 8 \\ 1 \\ 2 \end{pmatrix}$ $\therefore Q \left(\frac{8}{11}, \frac{1}{11}, \frac{2}{11} \right)$ $l_{RS} : \mathbf{r} = \begin{pmatrix} -5 \\ -6 \\ -7 \end{pmatrix} + \mu \begin{pmatrix} -2 \\ -3 \\ -6 \end{pmatrix}, \mu \in \mathbb{R} \text{ and } \pi_{base} : \mathbf{r} \cdot \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = -9$ To find R, $\left[\begin{pmatrix} -5 \\ -6 \\ -7 \end{pmatrix} + \mu \begin{pmatrix} -2 \\ -3 \\ -6 \end{pmatrix} \right] \cdot \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = -9$ $(-5-6-7) + \mu(-2-3-6) = -9 \Rightarrow \mu = -\frac{9}{11}$ $\therefore \mathbf{r} = \begin{pmatrix} -5 \\ -6 \\ -7 \end{pmatrix} - \frac{9}{11} \begin{pmatrix} -2 \\ -3 \\ -6 \end{pmatrix} = \frac{1}{11} \begin{pmatrix} -37 \\ -39 \\ -23 \end{pmatrix}$ $\therefore R \left(\frac{-37}{11}, \frac{-39}{11}, \frac{-23}{11} \right)$	Q and R are points of intersections between respective line and respective plane
ii	Since $\overrightarrow{PQ}$ is parallel to $\begin{pmatrix} 2 \\ 3 \\ 6 \end{pmatrix}$ and normal of surface parallel to $\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$, $\cos \theta = \frac{\begin{pmatrix} 2 \\ 3 \\ 6 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}}{\sqrt{2^2+3^2+6^2} \sqrt{1^2+1^2+1^2}} = \frac{11}{7\sqrt{3}}.$ $\overrightarrow{QR} = \frac{1}{11} \begin{pmatrix} -37 \\ -39 \\ -23 \end{pmatrix} - \frac{1}{11} \begin{pmatrix} 8 \\ 1 \\ 2 \end{pmatrix} = -\frac{5}{11} \begin{pmatrix} 9 \\ 8 \\ 5 \end{pmatrix}$	Read the question carefully, there is no need to solve for the angles. Since θ and β are acute, check that $\cos \theta$ and $\cos \beta$ are both positive.

	Since $\overrightarrow{PQ}$ is parallel to $\begin{pmatrix} 9 \\ 8 \\ 5 \end{pmatrix}$ and normal of surface parallel to $\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$, $\cos \beta = \frac{\begin{pmatrix} 9 \\ 8 \\ 5 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}}{\sqrt{170}\sqrt{3}} = \frac{22}{\sqrt{510}}.$	
iii	Thickness = $\overrightarrow{QR} \cdot \hat{n} = \left \frac{-5}{11} \begin{pmatrix} 9 \\ 8 \\ 5 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right \frac{1}{\sqrt{3}} = \frac{10}{\sqrt{3}}$	
iv	$\sin \theta = k \sin \beta$ $k = \frac{\sin \theta}{\sin \beta} = \frac{\sqrt{1 - \cos^2 \theta}}{\sqrt{1 - \cos^2 \beta}} = \frac{\sqrt{1 - \left(\frac{11}{7\sqrt{3}}\right)^2}}{\sqrt{1 - \left(\frac{22}{\sqrt{510}}\right)^2}} = 1.86 \text{ (to 3 s.f.)}$	$k = \frac{\sqrt{170}}{7}$
v	If $\beta > \theta$, then $\sin \beta > \sin \theta > 0$ since both β and θ are acute. Hence, $0 < k = \frac{\sin \theta}{\sin \beta} < 1$.	